Reality Selection Module (RSM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

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Governed Space

Reality Selection

Canonical Definition

Reality Selection Module (RSM) defines the governance conditions under which real-world entities, properties, behaviors, relationships, processes, events, environmental conditions, dependencies, constraints, disturbances, and other materially relevant reality elements are identified, evaluated, selected, excluded, prioritized, and documented for representation within a governed simulation.

RSM establishes the first transformation boundary between the reality relevant to a simulation inquiry and the subset of that reality that will actually enter the simulation representation.

Operational Role

RSM serves as the first internal governance space of the Reality Representation Standard (RRS).

The preceding Simulation Purpose & Scope Standard (SPSS) establishes why the simulation exists, what inquiry it must address, what remains outside its scope, the applicable context, and the required depth and precision.

RSM asks the next question:

Which elements of reality are materially relevant to that inquiry and therefore must be represented inside the simulation?

The governed progression is:

Simulation Purpose & Scope → Relevant Reality → Reality Element Identification → Materiality Assessment → Selection / Exclusion → Reality Selection Basis → Reality Abstraction

RSM does not determine how selected reality will technically be modeled.

It establishes what reality must first be carried forward into representation governance.

Module Operational Space

RSM governs: relevant reality identification, reality element identification, physical entity selection, environmental element selection, behavioral element selection, process selection, event selection, interaction selection, relationship selection, dependency selection, constraint selection, disturbance selection, temporal element selection, spatial element selection, operational condition selection, failure-condition selection, boundary-condition relevance, materiality assessment, selection priority, inclusion decisions, exclusion decisions, conditional inclusion, representation relevance, selection completeness, selection uncertainty, selection assumptions, selection dependencies, selection traceability, selection change, and Reality Selection Basis governance.

Module Function

RSM exists because no simulation represents all of reality.

Every simulation performs selection, whether explicitly or implicitly.

Without governed selection, materially important elements may disappear before model construction even begins.

This may create a simulation that is technically sophisticated but structurally incapable of answering its intended question.

RSM prevents: arbitrary reality selection, convenience-driven omission, selection based only on available data, selection based only on existing model capabilities, hidden exclusion of difficult phenomena, omission of material interactions, omission of relevant environmental conditions, omission of failure states, selection driven by desired simulation outcomes, unexplained reduction of real-world complexity, and later representation of an incomplete simulated world as if it represented the relevant reality comprehensively.

Reality Selection Principle

SIMULOS® establishes:

Reality should enter a simulation because it is material to the simulation inquiry, not merely because it is easy to model.

Technical availability does not determine methodological relevance.

A phenomenon may be difficult to represent and still be essential to the simulation.

Relevant Reality

Relevant Reality is the portion of reality that may materially affect the governed simulation inquiry.

It may include: physical objects, physical properties, system states, actors, behaviors, environmental conditions, interactions, dependencies, constraints, external influences, disturbances, failure mechanisms, temporal effects, spatial effects, uncertainty sources, and operational conditions.

Relevant Reality is established relative to the defined Simulation Object, Purpose, Scope, Applicability Context, and required Simulation Depth & Precision.

Reality Element

A Reality Element is an identifiable aspect of Relevant Reality considered for representation within the simulation.

A Reality Element may represent:

Entity

a physical, digital, biological, organizational, human, autonomous, or other actor or object.

Property

a characteristic relevant to simulation behavior.

State

a condition in which an entity or environment may exist.

Behavior

an action, reaction, response, or behavioral pattern.

Process

a sequence or transformation occurring within reality.

Interaction

a relationship through which two or more elements influence one another.

Dependency

a condition under which one element depends upon another.

Constraint

a real-world limitation affecting possible behavior.

Event

an occurrence capable of changing the relevant system state.

Environment

external conditions affecting the Simulation Object.

Disturbance

an external or internal influence capable of materially changing outcomes.

Reality Element Identification

Reality selection begins by identifying candidate Reality Elements.

Identification should not initially be constrained only by what the current simulation platform can represent.

The correct sequence is:

What matters?

then:

How can it be represented?

not:

What can our simulator already represent?

then:

That must be what matters.

Reality Materiality

A Reality Element is material where its presence, absence, behavior, variation, interaction, or incorrect representation could materially affect: simulation behavior, scenario progression, simulation outputs, interpretation, decision relevance, safety implications, operational conclusions, or the intended reliance on simulation results.

Materiality therefore connects reality selection directly to the purpose of the simulation.

Materiality Assessment

Each candidate Reality Element should be assessed against questions such as:

Could this element materially influence the Simulation Object?

Could its omission materially alter simulation behavior?

Could its variation materially change the result?

Could it interact with another selected element?

Could it become important under abnormal or failure conditions?

Could it affect the intended interpretation or use of simulation outputs?

Where the answer is materially yes, the element should not disappear from representation without an explicit governed decision.

Materiality Is Context-Dependent

The same reality element may be:

Material

for one simulation,

and:

Non-Material

for another.

Wind may be immaterial to one indoor robotic simulation but essential to an autonomous aerial system.

Human behavior may be irrelevant to a component stress simulation but critical to an evacuation simulation.

RSM therefore does not establish universal lists of mandatory reality elements.

It governs the method by which relevance is established.

Reality Selection Decision

A candidate Reality Element may receive a governed selection status such as:

Selected

The element must proceed into representation governance.

Conditionally Selected

The element becomes required under defined scenarios, states, or conditions.

Excluded — Non-Material

The element has been evaluated and determined not to materially affect the simulation inquiry.

Excluded — Outside Scope

The element belongs outside the approved Simulation Scope.

Representation Constraint Identified

The element is material but cannot currently be represented sufficiently.

Selection Undetermined

Available knowledge is insufficient to determine whether the element is material.

Selected Does Not Mean Fully Modeled

Selection establishes that an element matters.

It does not yet determine: abstraction, fidelity, resolution, model implementation, numerical technique, or technical representation.

Those questions belong to subsequent RRS governance and later SIMULOS® spaces.

Exclusion Governance

Exclusion is as important as inclusion.

A simulation may legitimately exclude large portions of reality.

However, exclusion should remain bounded by the simulation inquiry.

A material element should not be excluded merely because: data are unavailable, representation is expensive, computation is difficult, modeling is inconvenient, the simulation platform lacks capability, the phenomenon creates uncertainty, or its inclusion weakens the expected result.

Where such limitations exist, they should remain visible as representation limitations rather than silently becoming non-reality.

Explicit Exclusion

An explicit exclusion should identify:

what is excluded,

why it is excluded,

whether it is considered non-material or outside scope,

and:

whether the exclusion limits interpretation of simulation outputs.

Hidden Exclusion

A Hidden Exclusion occurs where a materially relevant Reality Element is absent from the simulation representation without explicit assessment or documentation.

Hidden exclusion is a significant simulation-governance problem because downstream users may not know that the simulated environment lacks an important part of the reality being examined.

Representation Constraint

A material Reality Element may be identified even when adequate representation is currently impossible.

RSM must preserve this distinction.

Therefore:

Cannot currently represent ≠ Not relevant

The resulting Representation Constraint should propagate into subsequent Reality Representation governance.

Interactions and Dependencies

Reality selection should not consider elements only individually.

Some elements become material because of their relationships.

A component may appear insignificant independently but become critical through: dependency, feedback, coupling, cascading effects, shared resources, synchronization, environmental interaction, or failure propagation.

RSM therefore governs both Reality Elements and material relationships between Reality Elements.

Normal and Abnormal Reality

Reality selection should not automatically represent only normal operating conditions.

Depending upon Simulation Purpose, Relevant Reality may include: nominal operation, degraded operation, edge conditions, rare events, failure states, environmental extremes, conflicting actors, unexpected interactions, and recovery conditions.

Whether these must be selected depends upon the simulation inquiry.

Temporal Relevance

Some reality elements matter only over time.

RSM should therefore consider: duration, sequence, timing, latency, accumulation, degradation, adaptation, drift, and delayed consequences

where material to the simulation.

Spatial Relevance

Reality selection may also depend upon: location, distance, geometry, topology, proximity, physical boundaries, operational zones, or spatial relationships.

RSM identifies whether these characteristics are materially relevant.

Their later resolution belongs to representation governance.

Human Reality

Where humans materially interact with the Simulation Object, RSM may identify relevant: human behavior, reaction, decision-making, variability, error, communication, intervention, supervision, or coordination.

RSM does not govern human cognition itself.

Where cognitive governance is required, CLIA® may provide the complementary architecture.

Autonomous-System Reality

For autonomous systems, relevant reality may include: other autonomous actors, permissions, constraints, human intervention, environmental uncertainty, coordination, escalation conditions, failure states, communication loss, and unexpected actor behavior.

The governance of autonomous operation itself remains within ASGA™.

RSM governs only whether these real-world characteristics must be represented for the simulation inquiry.

Operational Reality Boundary

RSM does not establish authoritative Operational Reality.

ORA™ governs Operational Reality.

RSM consumes the relevant reality basis and determines which elements must enter simulation representation.

The distinction is:

ORA™ → What is the operational reality?

RSM → Which parts of that reality are material to this simulation?

Validation Boundary

RSM does not validate the simulation.

VALIDOS® governs validation.

RSM creates a governed Reality Selection Basis that may later become part of the evidence examined during validation.

Integrity Boundary

RSM does not establish general integrity governance.

INTEGROS® governs integrity.

RSM governs whether reality selection is explicit, bounded, traceable, and methodologically connected to the Simulation Purpose & Scope.

Reality Selection Completeness

Reality Selection Completeness asks whether the material categories of reality relevant to the simulation inquiry have been sufficiently considered.

Completeness does not mean representing everything.

It means:

no known materially relevant reality category has been omitted without governed treatment.

Selection Uncertainty

Sometimes it is unclear whether a Reality Element is material.

This uncertainty should remain visible.

Possible treatment may include: conditional selection, sensitivity analysis, scenario inclusion, additional investigation, conservative inclusion, or explicit unresolved status.

Uncertainty should not automatically become exclusion.

Selection Priority

Where representation resources are limited, selected Reality Elements may require prioritization.

Priority may reflect: materiality, consequence, sensitivity, uncertainty, dependency, failure significance, or relevance to the Simulation Purpose.

Prioritization does not remove the requirement to disclose material elements that cannot be represented sufficiently.

Reality Selection Traceability

Every material selection decision should remain traceable to the simulation inquiry.

A useful chain is:

Simulation Purpose → Simulation Scope → Relevant Reality → Reality Element → Materiality Assessment → Selection Decision → Representation Requirement

This allows downstream users to understand why a particular part of reality exists inside the simulation.

Reality Selection Change

Reality selection may need reassessment where: Simulation Purpose changes, Simulation Scope changes, the Simulation Object changes, operational reality changes, new dependencies are discovered, new failure modes emerge, new evidence becomes available, or simulation outputs reveal previously unrecognized sensitivity.

Selection Change Propagation

A material change in Reality Selection may affect downstream:

Abstraction → Fidelity → Resolution → Assumptions → Model & Environment → Inputs → Scenarios → Execution → Outputs → Reality Correlation

SIMULOS® therefore treats Reality Selection as an upstream governance dependency.

Reality Selection Basis

The principal governed output of RSM is the Reality Selection Basis.

It should establish: relevant reality context, candidate Reality Elements, materiality decisions, selected elements, conditional selections, excluded elements, exclusion rationale, unresolved selections, representation constraints, material interactions, material dependencies, and traceability to Simulation Purpose & Scope.

Minimum Implementation Framework

1. Establish Relevant Reality Context

Use the governed Simulation Object, Purpose, Scope, Applicability Context, and Depth & Precision requirements as the selection basis. 2. Identify Candidate Reality Elements

Identify potentially relevant: entities, properties, states, behaviors, processes, events, environments, interactions, dependencies, constraints, and disturbances.

3. Assess Materiality

Determine whether each candidate element could materially influence the simulation inquiry or interpretation. 4. Determine Selection Status

Classify each material candidate as: selected, conditionally selected, excluded with justification, representation-constrained, or undetermined.

5. Assess Relationships

Identify material interactions and dependencies between selected elements. 6. Assess Selection Completeness

Determine whether material categories of reality have been sufficiently considered. 7. Record Limitations and Uncertainty

Preserve unresolved materiality, unavailable knowledge, and representation constraints. 8. Establish Reality Selection Basis

Create the governed basis from which Reality Abstraction may proceed. 9. Preserve Traceability

Maintain:

Purpose → Scope → Reality → Materiality → Selection → Representation Requirement 10. Reassess After Material Change

Reopen Reality Selection when upstream simulation conditions or relevant reality materially change.

Governance Outputs

RSM may produce: Relevant Reality Context, Reality Element Register, Reality Element Classification, Reality Materiality Assessment, Reality Selection Decision, Selected Reality Element Set, Conditional Reality Selection Record, Reality Exclusion Record, Reality Exclusion Rationale, Representation Constraint Record, Reality Interaction Map, Reality Dependency Map, Reality Selection Uncertainty Record, Reality Selection Priority Record, Reality Selection Completeness Assessment, Reality Selection Change Record, Reality Selection Traceability Map, and Reality Selection Basis.

Use Case 1 — Autonomous Vehicle Simulation

An organization simulates an autonomous vehicle before deployment.

RSM prevents the simulated world from being defined merely by what the existing simulator already contains.

The organization evaluates whether the simulation must represent: other vehicles, pedestrians, cyclists, road geometry, weather, visibility, road degradation, emergency vehicles, unusual human behavior, sensor obstruction, communication failures, and infrastructure conditions.

A phenomenon that is materially relevant but difficult to simulate remains visible as a Representation Constraint rather than silently disappearing from the simulation. Use Case 2 — Spacecraft Landing Simulation

A spacecraft landing simulation requires selection of reality elements capable of materially influencing landing performance.

These may include: gravity, terrain, atmospheric conditions where applicable, vehicle dynamics, propulsion behavior, navigation uncertainty, sensor performance, communication delay, component failure, surface interaction, and environmental disturbances.

RSM establishes which of these must proceed into representation governance and which may legitimately remain outside the simulation inquiry.

The module does not determine the mathematical model used to represent them.

Architectural Position

Reality Selection is the first internal governance space of the Reality Representation Standard (RRS).

The complete RRS progression is:

Reality Selection → Reality Abstraction → Representation Fidelity → Representation Resolution → Representation Limitation

RSM establishes:

What parts of reality matter?

The next module, RAM — Reality Abstraction Module, establishes:

At what level of abstraction should those selected parts of reality be represented?

This separation is fundamental.

Selection determines what enters the simulated world.

Abstraction determines how much of its real-world complexity must remain once it enters.

Foundational Principle

A simulation cannot legitimately represent a reality element that was never recognized as relevant, and difficulty of representation does not make material reality irrelevant.

Canonical Closing Statement

Reality Selection Module establishes the first governed transformation from relevant reality into simulation representation by determining which entities, properties, states, behaviors, processes, events, environments, interactions, dependencies, constraints, disturbances, and other reality elements are material to the defined simulation inquiry. Through explicit materiality assessment, inclusion and exclusion governance, representation-constraint disclosure, relationship analysis, completeness assessment, uncertainty preservation, change propagation, and traceability to Simulation Purpose & Scope, RSM ensures that the simulated world is defined by what the simulation legitimately needs to examine rather than merely by what is convenient, available, or technically easy to model.

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